MeterMind — Model 2: GRU Seq2Seq with Attention

This notebook implements the **learned neural baseline** for MeterMind.

A GRU encoder-decoder with Bahdanau attention learns to compress expanded archaic prose back into metrically structured lines — purely from examples, no explicit stress rules.

How it works

```
INPUT  : 'Thou feedest thy light own flame with fuel that is self substantial'
ENCODER: reads each word left-to-right, producing a hidden state at each step
ATTENTION: at each decoder step, learns which encoder states to focus on
DECODER: generates one word at a time, attending to the right input words
OUTPUT : 'feedest thy lights flame with self substantial fuel'
```

Why attention?

Without attention, the encoder compresses the whole input into a single vector — a bottleneck. With attention, the decoder can look back at *all* encoder states at each step, learning which input word belongs at each output position. This maps naturally onto the reordering task.

Pipeline

```
training_pairs.csv → vocab → GRU+attention → train → evaluate (MA, SP, G)
```

0. Install dependencies

```
%pip install pronouncing sentence-transformers torch transformers --quiet

```

```
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 939.7/939.7 kB 19.2 MB/s eta 0:00:00
```

1. Imports & configuration

```
import re, io, csv, math, json, random
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader
from google.colab import files
from sentence_transformers import SentenceTransformer, util
import pronouncing

# ── Reproducibility ──────────────────────────────────────────────────────
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)

# ── Device ───────────────────────────────────────────────────────────────
# Automatically uses GPU if available (Colab provides a free T4)
DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Using device: {DEVICE}')

# ── Hyperparameters ──────────────────────────────────────────────────────
MAX_LEN    = 30     # max sequence length (words)
EMBED_DIM  = 128    # word embedding size
HIDDEN_DIM = 256    # GRU hidden state size
BATCH_SIZE = 32
EPOCHS     = 50
LR         = 0.001
```

```
CLIP       = 1.0    # gradient clipping to prevent exploding gradients
TEACHER_FORCING_RATIO = 0.5  # probability of using ground truth token during training

print('Config ready.')
```

```
Using device: cpu
Config ready.
```

2. Load dataset

```
uploaded = files.upload()
filename = list(uploaded.keys())[0]
df = pd.read_csv(io.StringIO(uploaded[filename].decode('utf-8')))

pairs = list(zip(df['input'].tolist(), df['target'].tolist()))
random.shuffle(pairs)

# Train / val / test split — split on unique targets so no target leaks across splits
unique_targets = sorted({t for _, t in pairs})
random.shuffle(unique_targets)
n = len(unique_targets)
train_targets = set(unique_targets[:int(n * 0.8)])
val_targets   = set(unique_targets[int(n * 0.8):int(n * 0.9)])
test_targets  = set(unique_targets[int(n * 0.9):])

train_pairs = [(s, t) for s, t in pairs if t in train_targets]
val_pairs   = [(s, t) for s, t in pairs if t in val_targets]
test_pairs  = [(s, t) for s, t in pairs if t in test_targets]

print(f'Total pairs : {len(pairs):,}')
print(f'Train       : {len(train_pairs):,}')
print(f'Val         : {len(val_pairs):,}')
print(f'Test        : {len(test_pairs):,}')
print(f'\nExample:')
print(f'  INPUT  : {pairs[0][0]}')
print(f'  TARGET : {pairs[0][1]}')
```

Choose Files training_pairs.csv

training_pairs.csv(text/csv) - 196446 bytes, last modified: 5/10/2026 - 100% done

```
Saving training_pairs.csv to training_pairs.csv
Total pairs : 1,455
Train       : 1,164
Val         : 144
Test        : 147

Example:
  INPUT  : Tan sacred beauty doth blunt the sharpest intents, that which is most hallowed in appearance.
  TARGET : Tan sacred beauty, blunt the sharpest intents,
```

3. Vocabulary

We build a word-level vocabulary from all training pairs. Four special tokens:

- `<PAD>` — padding to make sequences the same length in a batch
- `<SOS>` — start of sequence signal to the decoder
- `<EOS>` — end of sequence, tells decoder to stop
- `<UNK>` — unknown word (fallback for words not seen in training)

```
def tokenize(text):
    """Lowercase, strip punctuation, split into word tokens."""
    return re.sub(r"[^a-zA-Z\s]", "", text.lower()).split()


def build_vocab(pairs):
    vocab = {'<PAD>': 0, '<SOS>': 1, '<EOS>': 2, '<UNK>': 3}
    for src, tgt in pairs:
        for word in tokenize(src) + tokenize(tgt):
            if word not in vocab:
                vocab[word] = len(vocab)
    return vocab


```

```python
def encode(words, vocab, max_len):
    """Convert word list to padded integer sequence with SOS and EOS."""
    ids = [vocab['<SOS>']] + [vocab.get(w, vocab['<UNK>']) for w in words] + [vocab['<EOS>']]
    ids = ids[:max_len]
    return ids + [vocab['<PAD>']] * (max_len - len(ids))


# Build vocab from training pairs only (no data leakage)
vocab        = build_vocab(train_pairs)
idx_to_word  = {v: k for k, v in vocab.items()}
VOCAB_SIZE   = len(vocab)

print(f'Vocabulary size: {VOCAB_SIZE:,}')
```

```
Vocabulary size: 2,377
```

4. Dataset & DataLoader

```python
class PoetryDataset(Dataset):
    """
    Each item returns:
      src     : encoded input (expanded prose)
      tgt_in  : encoded target with SOS (decoder input)
      tgt_out : encoded target with EOS (what the decoder should predict)
    """
    def __init__(self, pairs, vocab, max_len):
        self.pairs   = pairs
        self.vocab   = vocab
        self.max_len = max_len

    def __len__(self):
        return len(self.pairs)

    def __getitem__(self, idx):
        src_text, tgt_text = self.pairs[idx]
        src = encode(tokenize(src_text), self.vocab, self.max_len)
        tgt = encode(tokenize(tgt_text), self.vocab, self.max_len)
        src = torch.tensor(src, dtype=torch.long)
        tgt = torch.tensor(tgt, dtype=torch.long)
        return src, tgt[:-1], tgt[1:]   # tgt_in (SOS...), tgt_out (...EOS)


train_loader = DataLoader(PoetryDataset(train_pairs, vocab, MAX_LEN), batch_size=BATCH_SIZE, shuffle=True)
val_loader   = DataLoader(PoetryDataset(val_pairs,   vocab, MAX_LEN), batch_size=BATCH_SIZE)
test_loader  = DataLoader(PoetryDataset(test_pairs,  vocab, MAX_LEN), batch_size=BATCH_SIZE)

print(f'Train batches : {len(train_loader)}')
print(f'Val batches   : {len(val_loader)}')
print(f'Test batches  : {len(test_loader)}')
```

```
Train batches : 37
Val batches   : 5
Test batches  : 5
```

5. Model architecture

Encoder

A GRU that reads the input word by word, producing:

- `encoder_outputs` — hidden state at every time step (used by attention)
- `hidden` — final hidden state (used to initialise the decoder)

Attention (Bahdanau)

At each decoder step, attention computes a score between the current decoder hidden state and every encoder output. These scores are softmaxed into weights, then used to build a **context vector** — a weighted sum of encoder outputs. Intuitively: "given where I am in decoding, which input words should I focus on?"

Decoder

At each step the decoder takes:

1. The previous output word (or `<SOS>` at the start)
2. The context vector from attention
3. Its own previous hidden state ...and predicts the next output word.

```python
class Encoder(nn.Module):
    def __init__(self, vocab_size, embed_dim, hidden_dim):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, embed_dim, padding_idx=0)
        self.gru       = nn.GRU(embed_dim, hidden_dim, batch_first=True)
        self.dropout   = nn.Dropout(0.3)

    def forward(self, src):
        # src: [batch, seq_len]
        embedded = self.dropout(self.embedding(src))        # [batch, seq_len, embed_dim]
        outputs, hidden = self.gru(embedded)                # outputs: [batch, seq_len, hidden_dim]
        return outputs, hidden


class BahdanauAttention(nn.Module):
    """
    Computes attention weights over encoder outputs.
    score(decoder_hidden, encoder_output) = v * tanh(W1*decoder + W2*encoder)
    """
    def __init__(self, hidden_dim):
        super().__init__()
        self.W1 = nn.Linear(hidden_dim, hidden_dim)
        self.W2 = nn.Linear(hidden_dim, hidden_dim)
        self.v  = nn.Linear(hidden_dim, 1, bias=False)

    def forward(self, decoder_hidden, encoder_outputs):
        # decoder_hidden  : [batch, hidden_dim]
        # encoder_outputs : [batch, src_len, hidden_dim]
        dec = self.W1(decoder_hidden).unsqueeze(1)          # [batch, 1, hidden_dim]
        enc = self.W2(encoder_outputs)                      # [batch, src_len, hidden_dim]
        scores = self.v(torch.tanh(dec + enc)).squeeze(-1)  # [batch, src_len]
        weights = F.softmax(scores, dim=-1)                 # [batch, src_len]
        context = torch.bmm(weights.unsqueeze(1), encoder_outputs).squeeze(1)  # [batch, hidden_dim]
        return context, weights


class Decoder(nn.Module):
    def __init__(self, vocab_size, embed_dim, hidden_dim):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, embed_dim, padding_idx=0)
        self.attention = BahdanauAttention(hidden_dim)
        # Input to GRU = embedding + context vector
        self.gru       = nn.GRU(embed_dim + hidden_dim, hidden_dim, batch_first=True)
        self.fc_out    = nn.Linear(hidden_dim, vocab_size)
        self.dropout   = nn.Dropout(0.3)

    def forward(self, tgt_token, hidden, encoder_outputs):
        # tgt_token      : [batch] — single token at this step
        # hidden         : [1, batch, hidden_dim]
        # encoder_outputs: [batch, src_len, hidden_dim]
        embedded = self.dropout(self.embedding(tgt_token.unsqueeze(1)))  # [batch, 1, embed_dim]
        context, attn_weights = self.attention(hidden.squeeze(0), encoder_outputs)
        gru_input = torch.cat([embedded, context.unsqueeze(1)], dim=-1)  # [batch, 1, embed+hidden]
        output, hidden = self.gru(gru_input, hidden)                     # [batch, 1, hidden_dim]
        logits = self.fc_out(output.squeeze(1))                          # [batch, vocab_size]
        return logits, hidden, attn_weights


class Seq2SeqAttention(nn.Module):
    def __init__(self, vocab_size, embed_dim, hidden_dim):
        super().__init__()
        self.encoder = Encoder(vocab_size, embed_dim, hidden_dim)
        self.decoder = Decoder(vocab_size, embed_dim, hidden_dim)

    def forward(self, src, tgt, teacher_forcing_ratio=0.5):
        """
        Teacher forcing: during training, feed the ground-truth token as
        decoder input with probability `teacher_forcing_ratio`.
        This speeds up learning by keeping the decoder on track early in training.
        """
        batch_size, tgt_len = tgt.shape
```

```
        encoder_outputs, hidden = self.encoder(src)

        # First decoder input is always <SOS>
        dec_input = tgt[:, 0]
        all_logits = []

        for t in range(1, tgt_len):
            logits, hidden, _ = self.decoder(dec_input, hidden, encoder_outputs)
            all_logits.append(logits.unsqueeze(1))
            # Teacher forcing decision
            if random.random() < teacher_forcing_ratio:
                dec_input = tgt[:, t]           # use ground truth
            else:
                dec_input = logits.argmax(-1)   # use model prediction

        return torch.cat(all_logits, dim=1)  # [batch, tgt_len-1, vocab_size]


model     = Seq2SeqAttention(VOCAB_SIZE, EMBED_DIM, HIDDEN_DIM).to(DEVICE)
optimizer = torch.optim.Adam(model.parameters(), lr=LR)
criterion = nn.CrossEntropyLoss(ignore_index=0)  # ignore PAD tokens in loss

n_params = sum(p.numel() for p in model.parameters() if p.requires_grad)
print(f'Model parameters: {n_params:,}')
```

```
Model parameters: 2,140,745
```

6. Training

```
def run_epoch(model, loader, optimizer=None, teacher_forcing_ratio=0.5):
    """Run one epoch. If optimizer is None, runs in eval mode (no gradients)."""

    is_train = optimizer is not None

    if is_train:
        model.train()
    else:
        model.eval()

    total_loss = 0

    with torch.set_grad_enabled(is_train):

        for src, tgt_in, tgt_out in loader:

            src = src.to(DEVICE)
            tgt_in = tgt_in.to(DEVICE)
            tgt_out = tgt_out.to(DEVICE)

            logits = model(src, tgt_in, teacher_forcing_ratio)

            # ensure same sequence length
            min_len = min(logits.size(1), tgt_out.size(1))

            logits = logits[:, :min_len, :]
            tgt_out = tgt_out[:, :min_len]

            loss = criterion(
                logits.reshape(-1, VOCAB_SIZE),
                tgt_out.reshape(-1)
            )

            if is_train:
                optimizer.zero_grad()
                loss.backward()

                torch.nn.utils.clip_grad_norm_(
                    model.parameters(),
                    CLIP
                )

                optimizer.step()

            total_loss += loss.item()
```

```

    return total_loss / len(loader)
```

Save checkpoint

Download `best_model.pt` now so you don't lose it if the runtime resets.

```
# ── Save model checkpoint ──────────────────────────────────────────────
torch.save(model.state_dict(), 'best_model.pt')
files.download('best_model.pt')
print('Saved and downloaded best_model.pt')

```

```
Saved and downloaded best_model.pt
```

7. Inference

Two constraints applied at inference time:

1. **Copy mask** — model can only output words present in the source input (since all target words are guaranteed to be there by dataset construction)
2. **Repetition penalty** — reduces the score of already-used tokens so the model doesn't repeat words

```
# Load best checkpoint
model.load_state_dict(torch.load('best_model.pt', map_location=DEVICE))
model.eval()


def get_source_mask(src_ids, vocab_size):
    """Allow only tokens present in the source. Always allow EOS so model can stop."""
    mask = torch.full((vocab_size,), float('-inf'))
    for token_id in src_ids:
        if token_id not in (0, 1, 2, 3):  # skip PAD, SOS, EOS, UNK
            mask[token_id] = 0.0
    mask[vocab['<EOS>']] = 0.0
    return mask.to(DEVICE)


def predict(src_text, max_len=MAX_LEN, repeat_penalty=2.0):
    src_ids  = encode(tokenize(src_text), vocab, max_len)
    src      = torch.tensor(src_ids, dtype=torch.long).unsqueeze(0).to(DEVICE)
    src_mask = get_source_mask(src_ids, VOCAB_SIZE)

    with torch.no_grad():
        encoder_outputs, hidden = model.encoder(src)

    dec_input  = torch.tensor([vocab['<SOS>']], dtype=torch.long).to(DEVICE)
    output_words = []
    used_tokens  = set()

    for _ in range(max_len):
        with torch.no_grad():
            logits, hidden, _ = model.decoder(dec_input, hidden, encoder_outputs)

        logits = logits.squeeze(0) + src_mask
        for tok in used_tokens:
            logits[tok] -= repeat_penalty

        next_token = logits.argmax().item()
        if next_token == vocab['<EOS>']:
            break

        word = idx_to_word.get(next_token, '<UNK>')
        output_words.append(word)
        used_tokens.add(next_token)
        dec_input = torch.tensor([next_token], dtype=torch.long).to(DEVICE)

    return ' '.join(output_words)


# Quick sample
print('Sample predictions:\n')
```

```
for src, tgt in test_pairs[:3]:
    out = predict(src)
    print(f'INPUT  : {src}')
    print(f'TARGET : {tgt}')
    print(f'OUTPUT : {out}')
    print()


def predict_no_mask(src_text, repeat_penalty=2.0):
    """Inference WITHOUT copy mask — model can output any vocabulary word."""
    src_ids = encode(tokenize(src_text), vocab, MAX_LEN)
    src     = torch.tensor(src_ids, dtype=torch.long).unsqueeze(0).to(DEVICE)

    with torch.no_grad():
        encoder_outputs, hidden = model.encoder(src)

    dec_input    = torch.tensor([vocab['<SOS>']], dtype=torch.long).to(DEVICE)
    output_words = []
    used_tokens  = set()

    for _ in range(MAX_LEN):
        with torch.no_grad():
            logits, hidden, _ = model.decoder(dec_input, hidden, encoder_outputs)
        logits = logits.squeeze(0)
        # No source mask — model can generate any word in vocab
        for tok in used_tokens:
            logits[tok] -= repeat_penalty
        # Block special tokens only
        for special in [vocab['<PAD>'], vocab['<SOS>'], vocab['<UNK>']]:
            logits[special] = float('-inf')
        next_token = logits.argmax().item()
        if next_token == vocab['<EOS>']:
            break
        output_words.append(idx_to_word.get(next_token, '<UNK>'))
        used_tokens.add(next_token)
        dec_input = torch.tensor([next_token], dtype=torch.long).to(DEVICE)

    return ' '.join(output_words)

```

```
Sample predictions:

INPUT  : Which did erst canopy the herd from heat, as a protective shelter doth.
TARGET : Which erst from heat did canopy the herd,
OUTPUT : as a did from shelter erst

INPUT  : And right perfection, which hath been wrongfully disgraced by villainous hands
TARGET : And right perfection wrongfully disgraced,
OUTPUT : villainous

INPUT  : The weary night doth claim my heavy eyelids, and to it shall they fall?
TARGET : My heavy eyelids to the weary night?
OUTPUT : my and
```

8. Evaluation metrics

```
# ── Shared metric utilities ──────────────────────────────────────────────
# Upload meter_utils.py then eval_metrics.py when prompted.
from google.colab import files as _fu
print('Please upload meter_utils.py')
_fu.upload()
from meter_utils import *
print('Please upload eval_metrics.py')
_fu.upload()
from eval_metrics import *

# Trigger model downloads before the eval loop
load_sp_model()
load_gpt2()
print('All metrics ready.')

```

```
Please upload meter_utils.py
Choose Files meter_utils.py
meter_utils.py(n/a) - 5861 bytes, last modified: 5/10/2026 - 100% done
Saving meter_utils.py to meter_utils.py
Please upload eval_metrics.py
Choose Files eval_metrics.py
eval_metrics.py(n/a) - 3273 bytes, last modified: 5/10/2026 - 100% done
Saving eval_metrics.py to eval_metrics.py
Loading SentenceTransformer (first run downloads ~90 MB)...
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
modules.json: 100%  349/349 [00:00<00:00, 29.0kB/s]
config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 8.93kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
README.md: 10.5k/? [00:00<00:00, 860kB/s]
sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 4.38kB/s]
config.json: 100%  612/612 [00:00<00:00, 31.7kB/s]
model.safetensors: 100%  90.9M/90.9M [00:01<00:00, 456MB/s]
Loading weights: 100%  103/103 [00:00<00:00, 355.82it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100%  350/350 [00:00<00:00, 13.4kB/s]
vocab.txt: 232k/? [00:00<00:00, 10.9MB/s]
tokenizer.json: 466k/? [00:00<00:00, 18.4MB/s]
special_tokens_map.json: 100%  112/112 [00:00<00:00, 9.53kB/s]
config.json: 100%  190/190 [00:00<00:00, 12.3kB/s]
Loading GPT-2 (first run downloads ~500 MB)...
tokenizer_config.json: 100%  26.0/26.0 [00:00<00:00, 1.79kB/s]
vocab.json: 1.04M/? [00:00<00:00, 16.7MB/s]
merges.txt: 456k/? [00:00<00:00, 16.7MB/s]
tokenizer.json: 1.36M/? [00:00<00:00, 21.5MB/s]
config.json: 100%  665/665 [00:00<00:00, 59.7kB/s]
model.safetensors: 100%  548M/548M [00:08<00:00, 125MB/s]
Loading weights: 100%  148/148 [00:00<00:00, 405.57it/s, Materializing param=transformer.wte.weight]
GPT2LMHeadModel LOAD REPORT from: gpt2
Key                   | Status     |  |
----------------------+------------+--+-
h.{0...11}.attn.bias | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
generation_config.json: 100%  124/124 [00:00<00:00, 7.38kB/s]
All metrics ready.
```

9. Evaluate on test set

```
# ── Step 1 & 2: Inference + Metrical Accuracy ──────────────────────────────
from tqdm.notebook import tqdm

rows = []
for src_text, tgt_text in tqdm(test_pairs, desc='Inference (no mask) + MA'):
    output = predict_no_mask(src_text)
```

```
    words  = tokenize(output)
    rows.append({
        'input':  src_text,
        'target': tgt_text,
        'output': output,
        'MA':     round(metrical_accuracy(words), 4),
    })
print(f'{len(rows):,} predictions done.')

```

```
Inference (no mask) + MA: 100%                                    147/147 [00:07<00:00, 16.62it/s]
147 predictions done.
```

```
# ── Step 3: Semantic Preservation (batched encode) ──────────────────────────
from sentence_transformers import util

inputs_clean  = [' '.join(tokenize(r['input'])) for r in rows]
outputs_clean = [r['output'] for r in rows]

print(f'Encoding {len(rows):,} input+output pairs...')
embs = load_sp_model().encode(
    inputs_clean + outputs_clean,
    convert_to_tensor=True,
    show_progress_bar=True,
)

n = len(rows)
for i, r in enumerate(rows):
    r['SP'] = round(float(max(0.0, util.cos_sim(embs[i], embs[n + i]))), 4)

print('SP done.')

```

```
Encoding 147 input+output pairs...
Batches: 100%                                          10/10 [00:05<00:00,  2.43it/s]
SP done.
```

```
# ── Step 4: Grammaticality (batched GPT-2, per-sequence loss) ───────────────
import torch, math
from tqdm.notebook import tqdm as tqdm_nb

BATCH_SIZE = 32
gpt2_model, gpt2_tok = load_gpt2()

if gpt2_tok.pad_token_id is None:
    gpt2_tok.pad_token_id = gpt2_tok.eos_token_id

outputs_list = [r['output'] for r in rows]
g_scores     = []

for batch_start in tqdm_nb(range(0, len(outputs_list), BATCH_SIZE), desc='Grammaticality'):
    batch = outputs_list[batch_start : batch_start + BATCH_SIZE]
    enc   = gpt2_tok(batch, return_tensors='pt', padding=True,
                     truncation=True, max_length=128)
    input_ids = enc['input_ids']
    labels    = input_ids.clone()
    labels[input_ids == gpt2_tok.pad_token_id] = -100

    with torch.no_grad():
        logits = gpt2_model(**enc).logits

    shift_logits = logits[:, :-1, :].contiguous()
    shift_labels = labels[:, 1:].contiguous()
    loss_fn      = torch.nn.CrossEntropyLoss(ignore_index=-100, reduction='none')
    token_losses = loss_fn(
        shift_logits.view(-1, shift_logits.size(-1)),
        shift_labels.view(-1),
    ).view(len(batch), -1)
    mask       = (shift_labels != -100).float()
    seq_losses = (token_losses * mask).sum(-1) / mask.sum(-1).clamp(min=1)

    for loss_val in seq_losses.tolist():
        g_scores.append(round(1 / (1 + math.log(math.exp(loss_val))), 4))

for r, g in zip(rows, g_scores):
```

```
    r['G'] = g

print(f'G done. {len(g_scores):,} scores computed.')

```

```
Grammaticality: 100%                                                    5/5 [00:41<00:00,  7.23s/it]
G done. 147 scores computed.
```

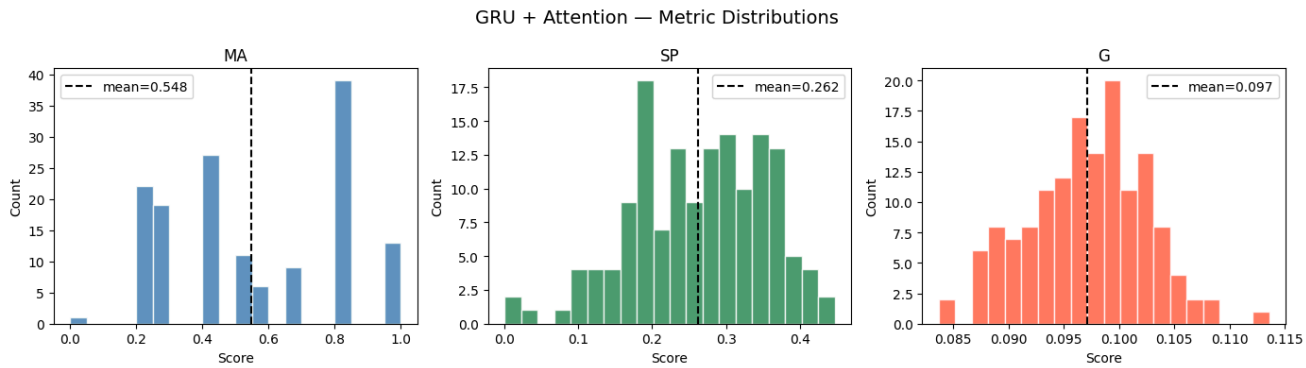
```
# ── Step 5: Assemble results DataFrame ──────────────────────────────────────
results_df = pd.DataFrame(rows)
print(f'Evaluated {len(results_df):,} test pairs.')

```

```
Evaluated 147 test pairs.
```

9b. Results

```
print('=== GRU + Attention — Test Set Results ===')
print(f"Metrical Accuracy (MA)    : {results_df['MA'].mean():.4f} ± {results_df['MA'].std():.4f}")
print(f"Semantic Preservation (SP): {results_df['SP'].mean():.4f} ± {results_df['SP'].std():.4f}")
print(f"Grammaticality (G)        : {results_df['G'].mean():.4f} ± {results_df['G'].std():.4f}")

print('\n=== Sample outputs ===')
for _, row in results_df.head(5).iterrows():
    print(f"INPUT  : {row['input']}")
    print(f"TARGET : {row['target']}")
    print(f"OUTPUT : {row['output']}")
    print(f"MA={row['MA']:.3f}  SP={row['SP']:.3f}  G={row['G']:.3f}")
    print()

# Distribution plots
fig, axes = plt.subplots(1, 3, figsize=(14, 4))
fig.suptitle('GRU + Attention — Metric Distributions', fontsize=14)
for ax, metric, colour in zip(axes, ['MA', 'SP', 'G'], ['steelblue', 'seagreen', 'tomato']):
    ax.hist(results_df[metric], bins=20, color=colour, edgecolor='white', alpha=0.85)
    ax.axvline(results_df[metric].mean(), color='black', linestyle='--', linewidth=1.5,
               label=f"mean={results_df[metric].mean():.3f}")
    ax.set_title(metric)
    ax.set_xlabel('Score')
    ax.set_ylabel('Count')
    ax.legend()
plt.tight_layout()
plt.savefig('gru_attention_metrics.png', dpi=150)
plt.show()
```

```
=== GRU + Attention — Test Set Results ===
Metrical Accuracy (MA)    : 0.5476 ± 0.2631
Semantic Preservation (SP): 0.2622 ± 0.0908
Grammaticality (G)        : 0.0971 ± 0.0054

=== Sample outputs ===
INPUT  : Which did erst canopy the herd from heat, as a protective shelter doth.
TARGET : Which erst from heat did canopy the herd,
OUTPUT : woman body gently observe otherwise perplex come fellow indeed tally hammered answered thy afterwards mysteries cease c
MA=0.200  SP=0.000  G=0.096

INPUT  : And right perfection, which hath been wrongfully disgraced by villainous hands
TARGET : And right perfection wrongfully disgraced,
OUTPUT : if spend miles speed seated summon courses witness begone mysteriously sharp theerise celestial worn reign nature presu
MA=0.400  SP=0.181  G=0.101

INPUT  : The weary night doth claim my heavy eyelids, and to it shall they fall?
TARGET : My heavy eyelids to the weary night?
OUTPUT : if spend restraining plainly pretense conjoined discovered living dwellsthat surely utterance mans chide muse instrumen
MA=0.800  SP=0.394  G=0.105

INPUT  : Time, spent in gentle numbers, passeth so idly hence, and is most foolishly consumed;
TARGET : In gentle numbers time so idly spent;
OUTPUT : if spend restraining plainly pretense conjoined discovered living dwellsthat surely utterance mans chide muse instrumen
MA=0.800  SP=0.358  G=0.108

INPUT  : Despise not the wrinkles that come upon thee, for this thy golden time doth shine with splendour despite of such blemis
TARGET : Despite of wrinkles this thy golden time.
OUTPUT : woman body gently observe robe movd page humble tongue savour tigers witnessed contrary separable visit thisif reduce p
MA=0.300  SP=0.271  G=0.100
```



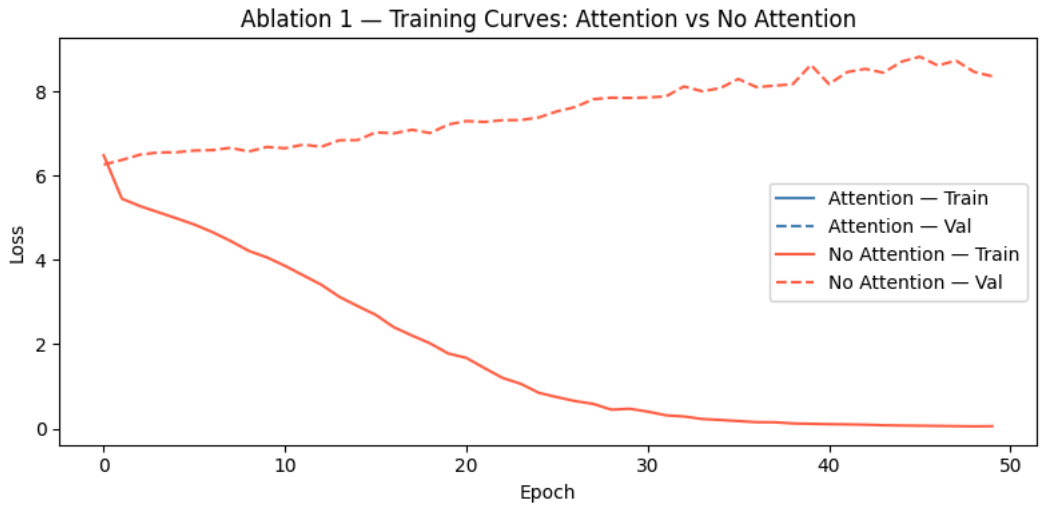
10. Ablation 1 — Attention vs No Attention

Train a plain GRU without attention and compare against the full attention model on the same test set.

```
# == ABLATION 1: Plain GRU (no attention) ==================================

class PlainDecoder(nn.Module):
    """Decoder without attention — uses only encoder final hidden state."""
    def __init__(self, vocab_size, embed_dim, hidden_dim):
        super().__init__()
        self.embedding = nn.Embedding(vocab_size, embed_dim, padding_idx=0)
        self.gru       = nn.GRU(embed_dim, hidden_dim, batch_first=True)
        self.fc_out    = nn.Linear(hidden_dim, vocab_size)
        self.dropout   = nn.Dropout(0.3)

    def forward(self, tgt_token, hidden, encoder_outputs):
        # encoder_outputs ignored — no attention
        embedded = self.dropout(self.embedding(tgt_token.unsqueeze(1)))
        output, hidden = self.gru(embedded, hidden)
        logits = self.fc_out(output.squeeze(1))
        # Return dummy weights (zeros) to keep interface consistent with Decoder
        dummy_weights = torch.zeros(
            tgt_token.size(0), encoder_outputs.size(1)
```

```
        ).to(tgt_token.device)
        return logits, hidden, dummy_weights


class PlainSeq2Seq(nn.Module):
    """GRU seq2seq WITHOUT attention — identical encoder, no-attention decoder."""
    def __init__(self, vocab_size, embed_dim, hidden_dim):
        super().__init__()
        self.encoder = Encoder(vocab_size, embed_dim, hidden_dim)
        self.decoder = PlainDecoder(vocab_size, embed_dim, hidden_dim)

    def forward(self, src, tgt, teacher_forcing_ratio=0.5):
        _, tgt_len = tgt.shape  # batch_size unused
        encoder_outputs, hidden = self.encoder(src)
        dec_input  = tgt[:, 0]
        all_logits = []
        for t in range(1, tgt_len):
            logits, hidden, _ = self.decoder(dec_input, hidden, encoder_outputs)
            all_logits.append(logits.unsqueeze(1))
            if random.random() < teacher_forcing_ratio:
                dec_input = tgt[:, t]
            else:
                dec_input = logits.argmax(-1)
        return torch.cat(all_logits, dim=1)


# —— Train plain model (reuses run_epoch from §6) ————————————————————————
plain_model     = PlainSeq2Seq(VOCAB_SIZE, EMBED_DIM, HIDDEN_DIM).to(DEVICE)
plain_optimizer = torch.optim.Adam(plain_model.parameters(), lr=LR)
plain_train_losses, plain_val_losses = [], []
best_plain_val = float('inf')

for epoch in range(1, EPOCHS + 1):
    train_loss = run_epoch(plain_model, train_loader, plain_optimizer, TEACHER_FORCING_RATIO)
    val_loss   = run_epoch(plain_model, val_loader)
    plain_train_losses.append(train_loss)
    plain_val_losses.append(val_loss)
    if val_loss < best_plain_val:
        best_plain_val = val_loss
        torch.save(plain_model.state_dict(), 'best_plain_model.pt')
    if epoch % 10 == 0:
        print(f'Epoch {epoch:3d}/{EPOCHS}  Train: {train_loss:.4f}  Val: {val_loss:.4f}')

print(f'Best plain val loss: {best_plain_val:.4f}')

# —— Training curve comparison ————————————————————————————————————————
plt.figure(figsize=(8, 4))
plt.plot(train_loss,       label='Attention — Train',    color='steelblue')
plt.plot(val_loss,         label='Attention — Val',      color='steelblue', linestyle='--')
plt.plot(plain_train_losses, label='No Attention — Train', color='tomato')
plt.plot(plain_val_losses,   label='No Attention — Val',   color='tomato',    linestyle='--')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.title('Ablation 1 — Training Curves: Attention vs No Attention')
plt.legend()
plt.tight_layout()
plt.savefig('ablation1_training_curves.png', dpi=150)
plt.show()

```

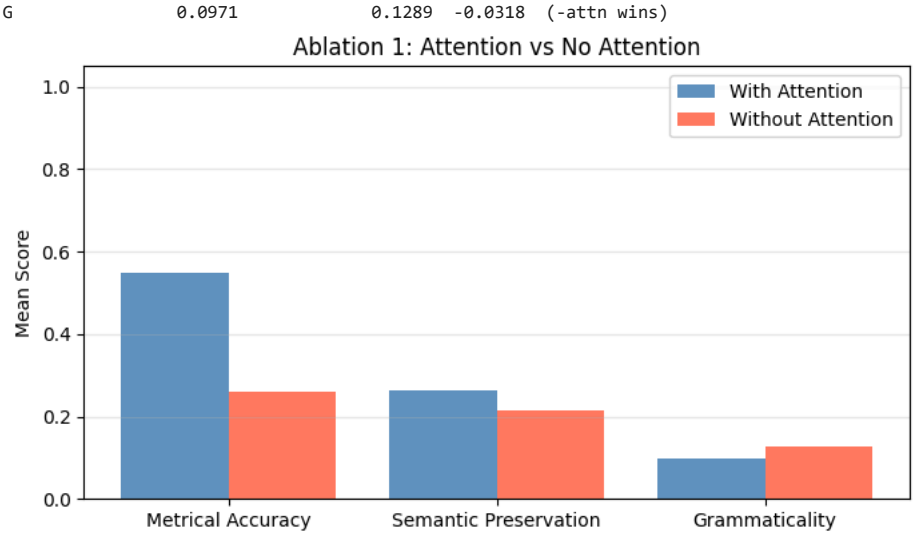
```
Epoch  10/50  Train: 4.0631  Val: 6.6837
Epoch  20/50  Train: 1.7818  Val: 7.2159
Epoch  30/50  Train: 0.4706  Val: 7.8480
Epoch  40/50  Train: 0.1132  Val: 8.6325
Epoch  50/50  Train: 0.0569  Val: 8.3611
Best plain val loss: 6.2650
```




```python
# ── Evaluate plain model on test set ──────────────────────────────────────
# Run after §9 so results_df and metric functions are already available.
plain_model.load_state_dict(torch.load('best_plain_model.pt', map_location=DEVICE))
plain_model.eval()

plain_results = []
for src_text, tgt_text in tqdm(test_pairs, desc='Evaluating plain GRU'):
    src_ids  = encode(tokenize(src_text), vocab, MAX_LEN)
    src      = torch.tensor(src_ids, dtype=torch.long).unsqueeze(0).to(DEVICE)
    src_mask = get_source_mask(src_ids, VOCAB_SIZE)

    with torch.no_grad():
        enc_out, hidden = plain_model.encoder(src)

    dec_input    = torch.tensor([vocab['<SOS>']], dtype=torch.long).to(DEVICE)
    output_words = []
    used_tokens  = set()

    for _ in range(MAX_LEN):
        with torch.no_grad():
            logits, hidden, _ = plain_model.decoder(dec_input, hidden, enc_out)
        logits = logits.squeeze(0) + src_mask
        for tok in used_tokens:
            logits[tok] -= 2.0
        next_token = logits.argmax().item()
        if next_token == vocab['<EOS>']:
            break
        output_words.append(idx_to_word.get(next_token, '<UNK>'))
        used_tokens.add(next_token)
        dec_input = torch.tensor([next_token], dtype=torch.long).to(DEVICE)

    output = ' '.join(output_words)
    words  = tokenize(output)
    plain_results.append({
        'input':  src_text,
        'target': tgt_text,
        'output': output,
        'MA': round(metrical_accuracy(words), 4),
        'SP': round(semantic_preservation(' '.join(tokenize(src_text)), output), 4),
        'G':  round(grammaticality(output), 4),
    })

plain_df = pd.DataFrame(plain_results)

# ── Side-by-side table ────────────────────────────────────────────────────
print('\n=== Ablation 1: Attention vs No Attention ===')
print(f"{'Metric':<6} {'With Attention':>16} {'Without Attention':>18} {'Delta':>8}")
print('-' * 52)
for metric in ['MA', 'SP', 'G']:
```

```
    with_att    = results_df[metric].mean()
    without_att = plain_df[metric].mean()
    delta       = with_att - without_att
    direction   = '(+attn wins)' if delta > 0 else '(-attn wins)'
    print(f"{metric:<6} {with_att:>16.4f} {without_att:>18.4f} {delta:>+8.4f}  {direction}")

# ── Bar chart ──────────────────────────────────────────────────────────
metrics      = ['MA', 'SP', 'G']
x            = range(len(metrics))
with_vals    = [results_df[m].mean() for m in metrics]
without_vals = [plain_df[m].mean()   for m in metrics]

fig, ax = plt.subplots(figsize=(7, 4))
ax.bar([i - 0.2 for i in x], with_vals,    0.4, label='With Attention',    color='steelblue', alpha=0.85)
ax.bar([i + 0.2 for i in x], without_vals, 0.4, label='Without Attention', color='tomato',    alpha=0.85)
ax.set_xticks(list(x))
ax.set_xticklabels(['Metrical Accuracy', 'Semantic Preservation', 'Grammaticality'])
ax.set_ylabel('Mean Score')
ax.set_ylim(0, 1.05)
ax.set_title('Ablation 1: Attention vs No Attention')
ax.legend()
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('ablation1_attention_comparison.png', dpi=150)
plt.show()

plain_df.to_csv('ablation1_no_attention_results.csv', index=False)
files.download('ablation1_no_attention_results.csv')
files.download('ablation1_training_curves.png')
files.download('ablation1_attention_comparison.png')
print('Ablation 1 complete.')

```

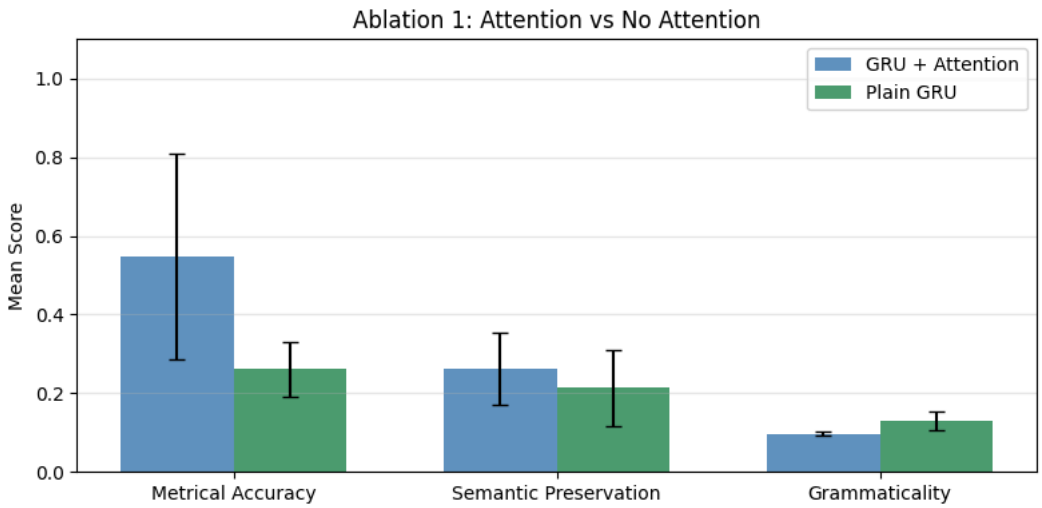
```
Evaluating plain GRU: 100%                                    147/147 [00:37<00:00,  4.35it/s]
`loss_type=None` was set in the config but it is unrecognized. Using the default loss: `ForCausalLMLoss`.

=== Ablation 1: Attention vs No Attention ===
Metric   With Attention  Without Attention    Delta
----------------------------------------------------
MA               0.5476             0.2612  +0.2864  (+attn wins)
SP               0.2622             0.2134  +0.0488  (+attn wins)
G                0.0971             0.1289  -0.0318  (-attn wins)
```




```
Ablation 1 complete.
```

```
# ── Ablation 1: Attention vs No Attention — comparison ──────────────────
import matplotlib.pyplot as plt
import numpy as np

plain_df = pd.DataFrame(plain_results)

metrics    = ['MA', 'SP', 'G']
attn_means = [results_df[m].mean() for m in metrics]
plain_means= [plain_df[m].mean()   for m in metrics]
attn_stds  = [results_df[m].std()  for m in metrics]
plain_stds = [plain_df[m].std()    for m in metrics]

```

```
x     = np.arange(len(metrics))
width = 0.35

fig, ax = plt.subplots(figsize=(8, 4))
ax.bar(x - width/2, attn_means,  width, yerr=attn_stds,  capsize=4,
       label='GRU + Attention', color='steelblue', alpha=0.85, ecolor='black')
ax.bar(x + width/2, plain_means, width, yerr=plain_stds, capsize=4,
       label='Plain GRU',       color='seagreen',  alpha=0.85, ecolor='black')
ax.set_xticks(x)
ax.set_xticklabels(['Metrical Accuracy', 'Semantic Preservation', 'Grammaticality'])
ax.set_ylabel('Mean Score')
ax.set_ylim(0, 1.1)
ax.set_title('Ablation 1: Attention vs No Attention')
ax.legend()
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('ablation1_attention.png', dpi=150)
plt.show()

print('\n=== Ablation 1: Attention vs No Attention ===')
print(f"{'Metric':<6} {'With Attention':>15} {'Without Attention':>18} {'Delta':>8}")
print('-' * 52)
for m, am, pm in zip(metrics, attn_means, plain_means):
    delta = am - pm
    winner = '(+attn wins)' if delta > 0 else '(-plain wins)'
    print(f"{m:<6} {am:>15.4f} {pm:>18.4f} {delta:>+8.4f}  {winner}")

files.download('ablation1_attention.png')

```

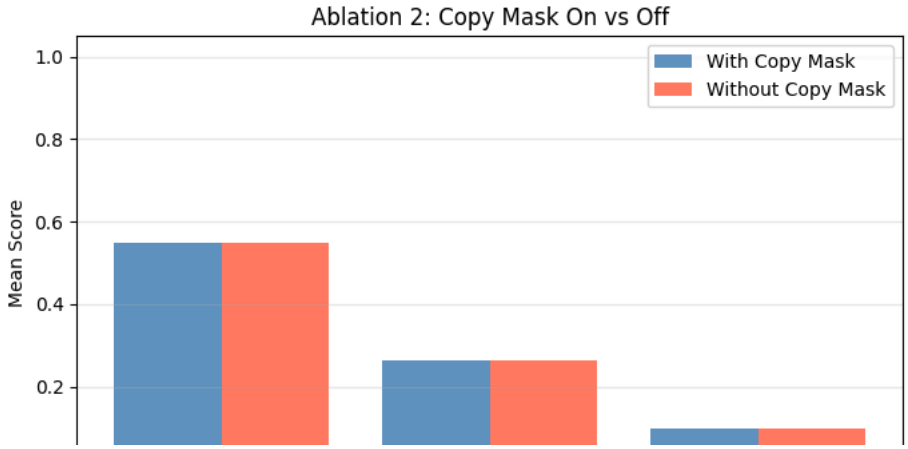



```
=== Ablation 1: Attention vs No Attention ===
Metric  With Attention  Without Attention    Delta
----------------------------------------------------
MA              0.5476             0.2612  +0.2864  (+attn wins)
SP              0.2622             0.2134  +0.0488  (+attn wins)
G               0.0971             0.1289  -0.0318  (-plain wins)
```

11. Ablation 2 — Copy Mask On vs Off

Re-run inference on the trained attention model with the copy mask disabled. No retraining needed — uses `results_df` as the with-mask baseline.

```
# == ABLATION 2: Copy Mask On vs Off ==========================================
# results_df (without copy mask) from §9 is the baseline.
# predict() with mask is run here to compare.


# — Run inference without mask ———————————————————————————————
no_mask_results = []
for src_text, tgt_text in tqdm(test_pairs, desc='Evaluating (no mask)'):
    output = predict_no_mask(src_text)
    words  = tokenize(output)
    no_mask_results.append({
        'input':  src_text,
```

```
        'input': src_text,
        'target': tgt_text,
        'output': output,
        'MA': round(metrical_accuracy(words), 4),
        'SP': round(semantic_preservation(' '.join(tokenize(src_text)), output), 4),
        'G':  round(grammaticality(output), 4),
    })

no_mask_df = pd.DataFrame(no_mask_results)

# ── Side-by-side table ──────────────────────────────────────────────────
print('\n=== Ablation 2: Copy Mask On vs Off ===')
print(f"{'Metric':<6} {'With Mask':>12} {'Without Mask':>14} {'Delta':>8}")
print('-' * 44)
for metric in ['MA', 'SP', 'G']:
    with_mask    = results_df[metric].mean()
    without_mask = no_mask_df[metric].mean()
    delta        = with_mask - without_mask
    direction    = '(+mask wins)' if delta > 0 else '(-mask wins)'
    print(f"{metric:<6} {with_mask:>12.4f} {without_mask:>14.4f} {delta:>+8.4f}  {direction}")

# ── Bar chart ───────────────────────────────────────────────────────────
metrics      = ['MA', 'SP', 'G']
x            = range(len(metrics))
with_vals    = [results_df[m].mean()  for m in metrics]
without_vals = [no_mask_df[m].mean()  for m in metrics]

fig, ax = plt.subplots(figsize=(7, 4))
ax.bar([i - 0.2 for i in x], with_vals,    0.4, label='With Copy Mask',    color='steelblue', alpha=0.85)
ax.bar([i + 0.2 for i in x], without_vals, 0.4, label='Without Copy Mask', color='tomato',    alpha=0.85)
ax.set_xticks(list(x))
ax.set_xticklabels(['Metrical Accuracy', 'Semantic Preservation', 'Grammaticality'])
ax.set_ylabel('Mean Score')
ax.set_ylim(0, 1.05)
ax.set_title('Ablation 2: Copy Mask On vs Off')
ax.legend()
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('ablation2_copy_mask.png', dpi=150)
plt.show()

no_mask_df.to_csv('ablation2_no_mask_results.csv', index=False)
files.download('ablation2_no_mask_results.csv')
files.download('ablation2_copy_mask.png')
print('Ablation 2 complete.')

```

```
Evaluating (no mask): 100%                                    147/147 [01:09<00:00,  2.54it/s]

=== Ablation 2: Copy Mask On vs Off ===
Metric    With Mask   Without Mask    Delta
--------------------------------------------
MA           0.5476         0.5476  +0.0000  (-mask wins)
SP           0.2622         0.2622  +0.0000  (-mask wins)
G            0.0971         0.0971  +0.0000  (-mask wins)
```